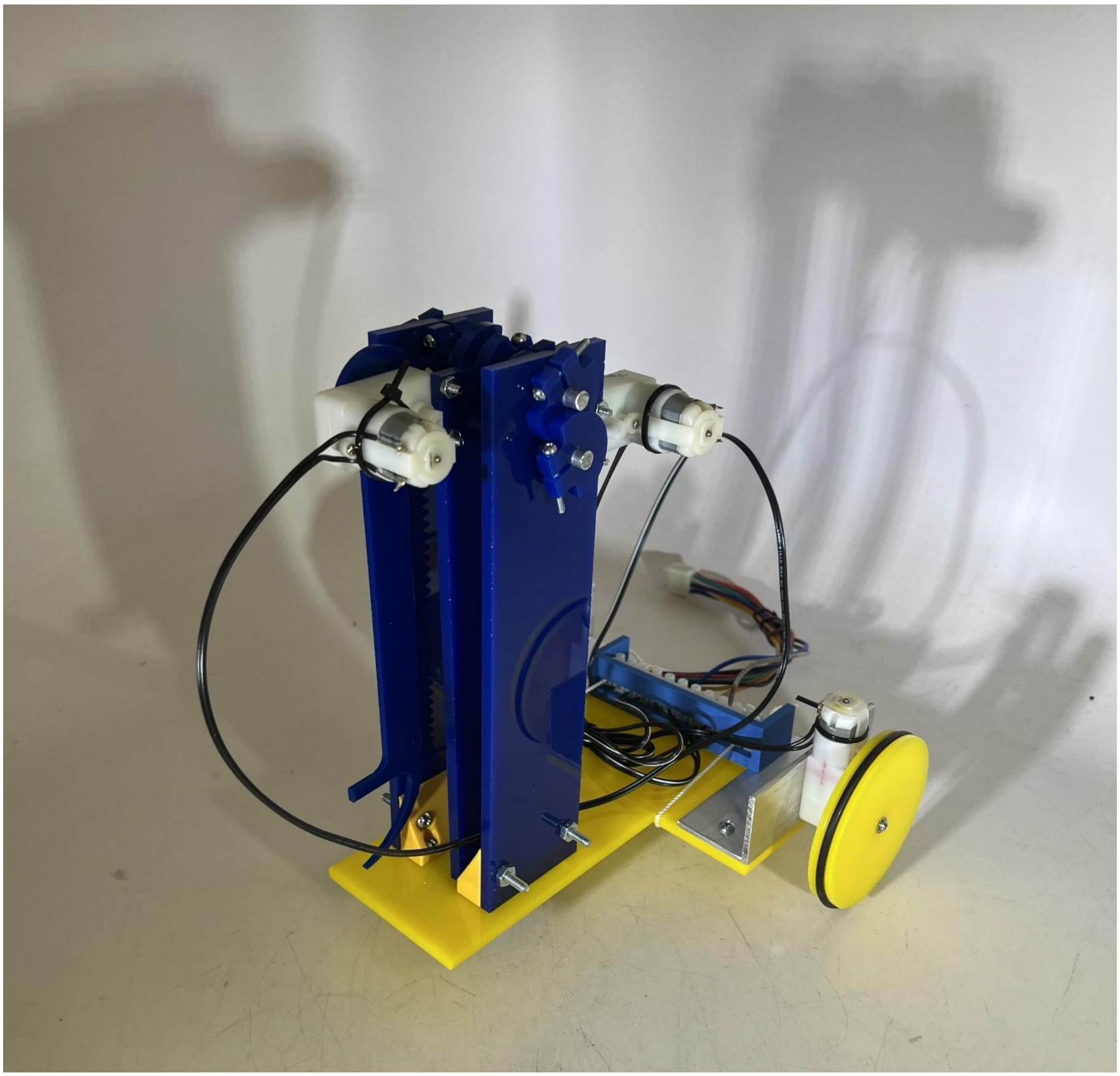
MAE 3 Robot Final Report

Team 32: Shaq and Onion

By: Jon Miranda



Part I: Description of Component

My team's robot has three powered components that make it a viable competitor for this competition. Those three components are the drivetrain, rack and pinion, and a rotating hook.

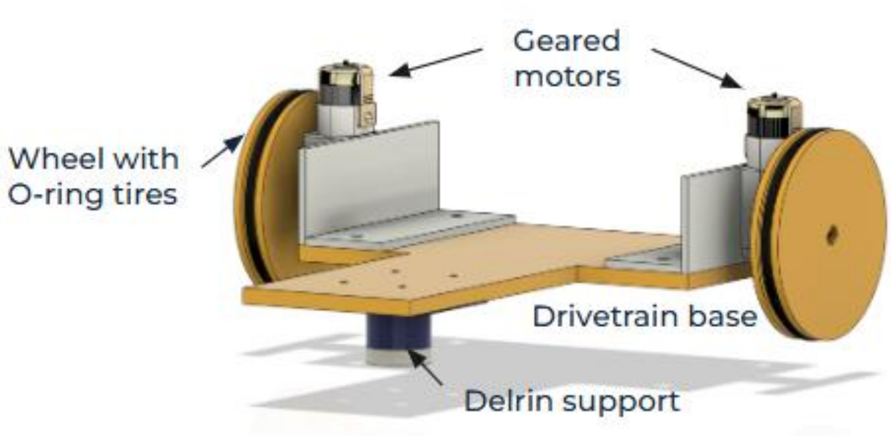


Figure 1: Drivetrain

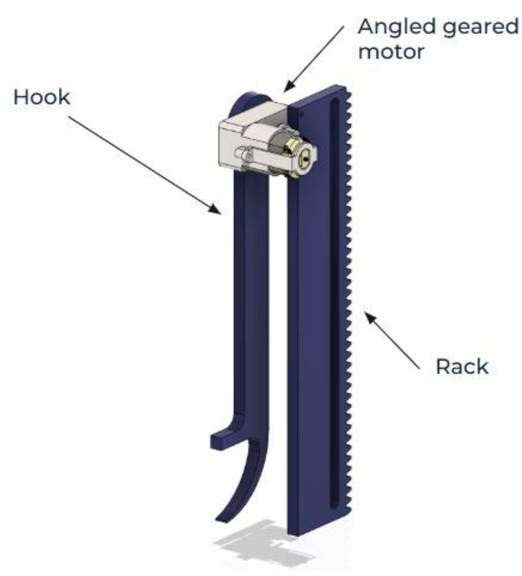


Figure 2: Rotating hook attached to rack

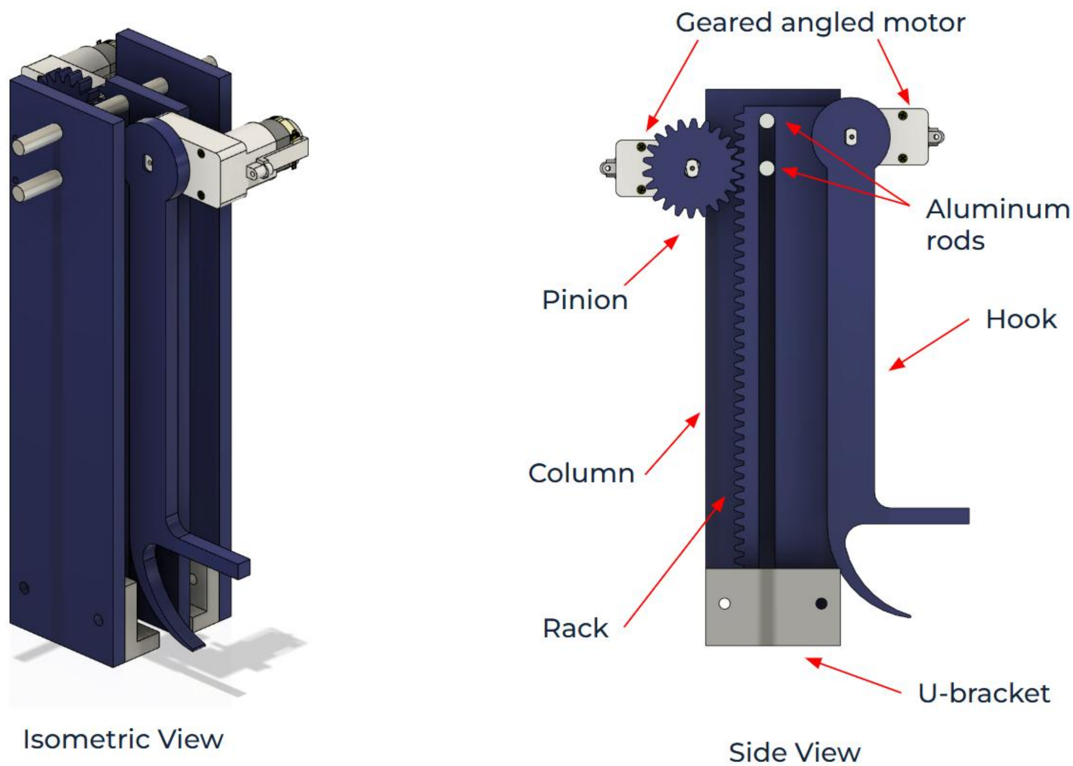


Figure 3: Component that will be analyzed (Rack and Pinion)

The drivetrain is controlled by two geared (straight motors) on each wheel allowing it to be able to rotate its orientation without the requirement of going forwards or backwards. Like how a roller works, the Delrin is a material with low friction, especially on the flooring of the competition stadium allowing it to rotate with ease, despite most of the weight being towards the front. The rotating hook has the freedom of rotating just under 270 degrees and this is the main device that will physically pick up and drop the rings in the desired spots. The rack and pinion, arguably the most important component is what allows the robot to reach new heights (quite literally) as the hook is attached to the rack and the rack moves up and down allowing the hook to go low to pick up rings, but also go high to drop it on the higher arms of the robot as those give an incentive of higher point multipliers. Due to the only having one minute to pick up and score as many rings for this competition, as a team, we came up with some requirements for our robot.

Minimum set of requirements:

- Be able to rotate $180°$ in 2 seconds
- Be able to lift the rack in 3 seconds
- Be able to rotate the hook $180°$ in 2 seconds
- Pinion must be able to lift the rack, hook, motor, and at least one ring ($mass = 104\ g$)

Each component had operated the way we expected too and even exceeded some expectations. Despite the amount of weight that the rack has and the lack of gear ratios, the rack was able to move from the bottom position to the highest position in about 2 seconds, even with 2 rings on it. This also means that the pinion gear was lift $112g$ instead of only $104g$. There was a slight trade off with the hook. There was no gear ratio for the hook, so the speed ended up being quite high due to its long length. It was able to rotate $270°$ in about two seconds, but there was a slight lack of control, and the torque was insufficient as if the moment was too strong (when the hook is completely perpendicular to the rack), sometimes the weight of multiple rings would cause it fall straight down. The only component that met our expectations was our drivetrain setup which rotated roughly $180°$ in two seconds, even with most of the weight being on the opposite end of the wheels. These values make perfect sense, although the speed of the rack barely worked out and this makes sense due to the amount of weight and lack of gear ratios. While the speed and lack of control of the hook was confusing, we first thought that it was a weight issue, so as a team, we tried making the hook thicker, but the issue remained the same. After further analysis and thinking, we realized that not utilizing a gear ratio and the long length of the hook led to a fast, but weak design. Despite these setbacks, our robot had an average score of 900 and an max score of 1200 meaning that we performed pretty solid.

Part II: Analysis of Component

One of the key components of my team's robot is the speed of the rack on the rack and pinion set up as the vertical movement of my robot is what allows the robot to pick up the rings and successfully place them on the miniature Sun God statue. The speed of the rack along is what determines the number of trips we can make to put multiple rings on the statue. To determine the potential of my robot, I will be determining how fast the rack can move when it is holding two rings since my team's strategy is to pick up two rings at a time.

Power and Energy Analysis:

To make this analysis feasible, there were a few assumptions that I made. My analysis is based on a rack that moves perfectly with no skipping, the motor will be running at maximum power, the pitch teeth have a pressure angle of 20 degrees and for the physics and mathematical portion, I will be doing a quasi-static analysis. While I am considering the angle of pressure, I will be neglecting the effects of friction between the gear teeth for the sake of simplicity. The hook can move just under 270°, but the biggest challenge is moving the hook when it is completely perpendicular to the rack, so I will be analyzing this system with the hook is perpendicular to the rack creating a larger moment. The values that were given were measured ourselves as we did lots of dimensioning and an FTPE analysis for all the parts included in our robot kit.

Measured values:

$$
\begin{aligned}
m_m &= 0.032\ kg \\
m_p &= 0.007\ kg \\
m_H &= 0.028\ kg \\
m_{rack} &= 0.050\ kg \\
m_{ring} &= 0.008\ kg \\
\tau_{stall} &= 0.28\ Nm \\
\omega_{no\ load} &= 8.00\ rad/sec \\
\theta_{pressure} &= 20° \\
\mu_{arcylic-aluminum} &= 0.3
\end{aligned}
$$

$$
\begin{aligned}
r_p &= 0.0183388m \\
h &= 0.17272m \\
d_1 &= 0.027m \\
d_2 &= 0.083m \\
d_3 &= 0.164m \\
d4 &= 0.0191m \\
d_5 &= 0.056m \\
d_6 &= 0.059m
\end{aligned}
$$

For this analysis, I will be drawing to free body diagrams, one for the pinion attached to the motor and another for rack, hook, other motor, and two rings. While this is analysis on the speed of the rack, I wanted to include the other components as those cause a moment, and the rack is carrying all that weight while moving up and down.

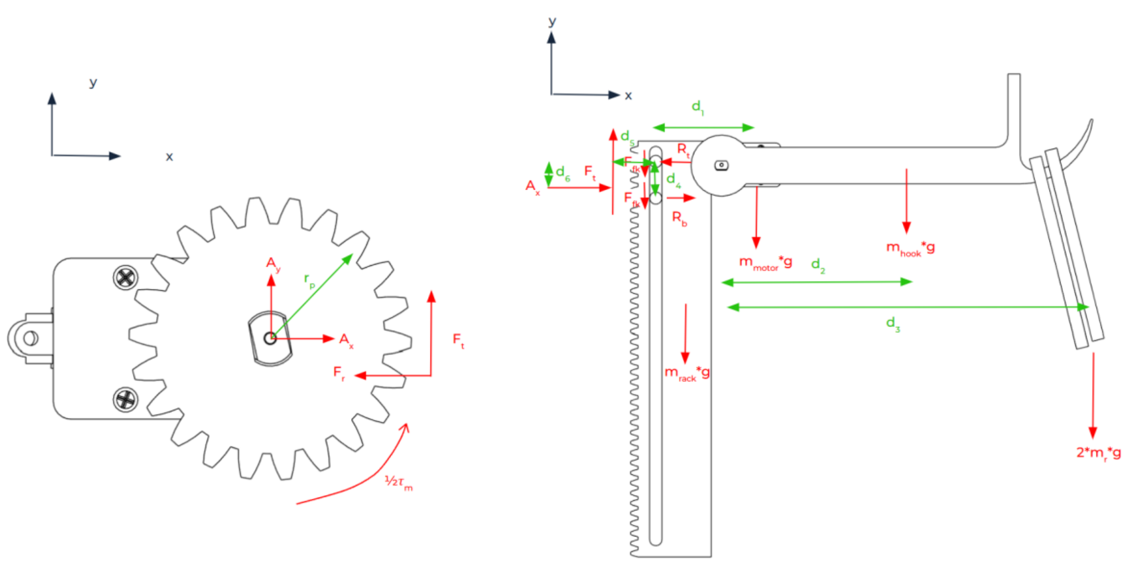


Figure 4 & 5: Free Body Diagram of Rack and other components and a Free Body Diagram for the pinion gear directly attached to a geared motor

As we can see from the free body diagrams, there are reaction forces on the support of the pin. This reaction force adds another force onto the rack which I will calculate below. The main thing from this analysis are the moments due to the forces of gravity on the individual parts of our rack and pinion. It's also important to consider the mass of the rack as we are also lifting the rack with the entire system to raise the hook.

Calculations:

1. Calculate the reaction force Ax as this exerts a force onto the rack causing another moment.
   (We only use half of the stall torque because we are assuming the motor is operating at maximum power)

$$Ft = \frac{(\frac{1}{2} * \tau_{stall})}{r_p}$$

$$Fr = F_t \times \tan{(20°)}$$

$$\sum Fx = A_x - F_r = 0$$

$$A_x = F_r$$

2. Calculate R_b (reaction force from bottom rod) by taking the moment about the top rod.

$$\sum M_t = R_b \times d_4 - m_m \times g - m_h \times (d_1 + d_2) \times g - 2 \times m_{ring} \times g \times (d_1 + d_3)$$
$$- F_t \times d_5 + A_x \times d_6 = 0$$

$$R_b = \frac{(m_m \times g + m_h \times (d_1 + d_2) \times g + 2 \times m_{ring} \times g \times (d_1 + d_3) + F_t \times d_5 - A_x \times d_6)}{d_4}$$

3. Calculate the reaction force at the top rod using sum of forces in the x direction.

$$\sum F_x = R_b + A_x - R_t = 0$$

$$R_t = R_b + A_x$$

4. Calculate the work due to friction and the work due to gravity

$$W_{friction} = R_t \times \mu_{arcylic-aluminum} \times h + R_b \times \mu_{arcylic-aluminum} \times h$$

$$W_{gravity} = \left(m_{motor} + m_{hook} + 2 \times m_{ring} + m_{rack}\right) \times g \times h$$

$$W = W_{friction} + W_{gravity}$$

5. Calculate how much time it takes for the motor to fight through all the work

$$Energy = P \times \Delta t$$

$$\Delta t = \frac{W}{P}$$

6. Calculate the average velocity by dividing the change in height by the time elapsed.

$$v_{rack} = \frac{h}{\Delta t}$$

How accurate was this analysis?

After doing all of these calculations, the final velocity of the rack came out to $v_{rack} = 1.91 \frac{inches}{second}$. I did an experiment where I had my teammates put two rings on the end of the hook just like my free body diagram and had them go from the lowest position to the highest position in one movement. After taking a video and doing careful analysis, I got that the actual velocity of the rack when carrying two rings is $3.4 \frac{inches}{second}$ giving me a percent error of 43.71%. While this seems bad, this gives me a safety factor of 1.78 which isn't the best, but it still means we have wiggle room based off my analysis. So, what can I conclude from this analysis? Was my analysis correct at all? My analysis had many general solutions, but the method of it was correct, at least from my perspective. I got a realistic number and used values that were personally measured by my team and the MAE 3 staff. I believe that I can see where the discrepancy in my analysis is. My analysis assumed that the aluminum rods and acrylic plastic were making 100% contact, but they weren't. For one, I had designed the slider in the rack to be a little bit bigger so that I was small enough to keep it stable, but wide enough so that it could slide nicely. Also, while the laser cutter is pretty price, the laser cutting isn't 100% accurate or precise meaning that the holes have a different size than what I asked for, and based off my analysis, I think slider hole came out just a bit bigger also allowing for much less friction. The design choice we took is limited in a sense where we can't change the speed or strength of the movement of the rack because it is spaced for the specific gear size for the pinon. This means that if we aren't satisfied with the output of the rack velocity, there's not much we can do besides modifying our entire design, which isn't realistic this far into the project. If I were given a second chance, I would most likely add a gear ratio that I would personally calculate before even doing a CAD due to the specific spacing we need. I wouldn't change the use of the rack and pinion because it's simple. Compared to our competitors, it seems lack luster, but towards the end of the intersection competition, we weren't the lowest scoring team. The biggest technical lesson I pulled from this was the importance of gear ratios and I know that next time, I should try to implement them, especially in higher level robotics since I am specializing in controls and robotics. Despite the limitations and simplicity, for the first major project I've done with a team, especially being heavily involved, I'm satisfied with how it performs in competition.

Part III: Design Process Essay

A major conceptual block I had during this design process was towards the end of the manufacturing period as we had everything created, but when it came to alignment, nothing had worked properly. Our team had made most of our components out of sheet metal which we thought was a good idea, but I realized our main issue was lack of precision. That's when I realized that we should make more of our parts out of plastic as the 3D printing and laser cutting allow for more precision as the dimensions are handled by a computer, making it less prone to human error. I consulted my team and asked if we could remake everything into plastic. The columns you see were originally sheet metal and the u-bracket was originally two L-brackets on the outside of the rack and pinon assembly.

The component for testing we chose was the rack and pinion and this proved to be the most challenging and most important component of our robot. We built a prototype rack and pinion out of foamcore, scrap acrylic, and hot glue. Our design had worked very reliably and was able to carry about $386g$ before failing. Using a rack and pinion for testing also taught us the importance of alignment and the way we mounted the motor was also something that was very feasible. The success of our risk reduction is what led us to use it in the final design of our robot.

While the Gantt chart seemed tedious and annoying at first, it became very useful and it was easy for me to manage my tasks, even while taking two other MAE courses, one of them being upper division. During the risk reduction period and the time of the required working component, we prioritized straight manufacturing without drawing it on CAD first.

While this worked out temporarily, I argue that this led to many of the alignment issues and this lack of planning led to us scrapping designs altogether. This led to us making late design changes which we learned in lecture is the most expensive change to do when it comes to design. This is when the Gantt chart came in handy. We prioritized the drive train because that was the easiest component to make, and it would set the foundation for the rest of our robot. During the entire rebuilding process, I decided to spend a few days of no manufacturing, but strict CAD modeling which included all the individual parts and even the entire robot. When we manufactured this time around, there wasn’t nearly as much trial, and error and our robot was working within 30 minutes to an hour.

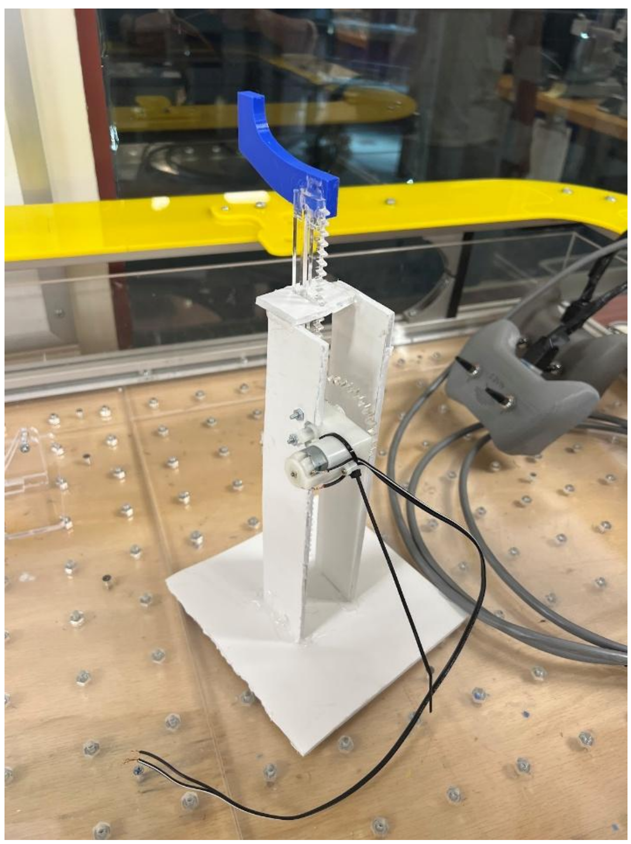


Figure 6: Prototype Risk Reduction